Francis Phan

Senior Software Engineer, Infrastructure & Reliability

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SUMMARY

Infrastructure-minded senior engineer with 6 years at AWS running the Amazon EFS control plane across 15+ regions, and the go-to incident lead for high-severity outages. Since 2025, the sole engineer building integrations, internal tools and AI services for a luxury hospitality business. Works in Python, Java and TypeScript on AWS and Linux.

SKILLS

Cloud & Infra	AWS (EC2, DynamoDB, SQS, SNS, Lambda, S3, IAM, CloudWatch, CloudFormation, CDK, Secrets Manager), Linux, shell scripting, infrastructure as code
Reliability	On-call and incident command, observability and roll-up alarming, runbooks, COEs/postmortems, emergency multi-region deployments
Delivery	CI/CD (GitHub Actions, Amazon Pipelines/Apollo), automated deploys, security reviews
Languages	Python, Java, TypeScript/JavaScript, Ruby, SQL
AI in production	MCP servers, Claude agent loops, usage-driven reliability tuning

EXPERIENCE

Systems & Software Consultant

2025 – Present

The Vines of Mendoza · Remote / Uco Valley, Argentina

- Sole engineer for infrastructure, integrations and internal tools at a vineyard resort and ownership community.
- Built a hotel-to-Salesforce guest sync that polls the on-prem Oracle database; runs unattended with automated deploys and a pause-and-replay safeguard.
- Replaced an estimated ~$14K-a-year Entrepreneurial Operating System (EOS) subscription with an in-house app costing under $2K a year, extending it to all ~100 staff in English and Spanish.
- Built an MCP server unifying Salesforce and NetSuite data so sales, marketing and client services can find information and produce reports easily; weekly usage reports drive continuous improvement, cutting the tool error rate from ~24% to ~10% and degraded calls to zero.
- Built an AI Slack assistant that lets staff look up customer data through the MCP server, and automatically builds guest profiles from public sources so hotel, resort and restaurant staff can tailor each stay.

Software Development Engineer

2018 – 2024

Amazon Web Services — EFS and EBS · Boston, MA

- Developed and operated the EFS Control Plane, the customer-facing API managing file system resources across 15+ regions.
- Incident lead for high-severity outages, from internal failures to dependencies like Route 53 and DynamoDB: ran war rooms, tracked recovery, and identified and notified affected customers, including a 12+ hour Route 53 outage; wrote and drove COEs.
- Automated on-call operations with scripts for bulk resource recovery and customer impact assessment.
- Built roll-up alarms for dependency outages like Route 53 and DynamoDB, cutting hundreds of pages to a handful so on-call could fight the outage, not the pager.
- Led the Log4Shell emergency response for EBS Fast Snapshot Restore: patched multiple services in every region, including China and GovCloud, in ~24 hours, then wrote the emergency-deploy runbook.
- Proposed and led a ~6-person rewrite of the EFS console off a deprecated stack, speeding up feature releases, cutting new-hire setup from weeks to ~1 hour, and passing an external pentest and AppSec review.
- Mentored interns and new hires, including hosting a summer intern on the console rewrite.
- Shipped IAM resource policy support for EFS, and maintained CI/CD pipelines, CloudFormation infrastructure, and CloudWatch dashboards and alarms.

EDUCATION

B.S. in Computer Science, Minor in Mathematics · University of Massachusetts Amherst **2014 – 2017**

- Lead web developer: led a Ruby on Rails team building an internal app for IT classroom consultants.